Abdul Hadi Jalil

Data Engineer | Applied AI and Flutter Development

Lahore, Pakistan | abdulhadijalil34@gmail.com | LinkedIn | GitHub | Portfolio

Professional Profile

Data Engineer at DotLabs with experience building production ETL and ELT pipelines in Python and SQL on AWS and Azure. Develops document AI and machine learning services with FastAPI and builds Flutter apps that connect those services to users.

Experience

Data Engineer | DotLabs | Dec 2025 to Present

- Built production ETL and ELT pipelines for water quality, industrial and commodities analytics. Managed ingestion, orchestration and PostgreSQL data quality checks using Airflow, Databricks and cron.
- Built pipelines for INCA model runs and hydrological and weather data across 10+ catchments. Loaded S3 data into PostgreSQL and supported the AppSync and Lambda GraphQL API.
- Integrated Claude tool use through MCP for natural language database queries and built a RAG workflow over documentation for employee onboarding.
- Worked with Azure Functions, Bicep and managed identities for cloud data workloads.

Python Developer | CodSoft | Jul to Aug 2023

- Built Python utilities including a weather API client and a task list, and used pandas to process small datasets.

Technical Skills

Data engineering | Python, SQL, pandas, Airflow, PySpark, Databricks, PostgreSQL, ETL/ELT, data validation

Cloud and APIs | AWS S3, RDS, Lambda, EC2 and AppSync, Azure Functions, Bicep, managed identity, FastAPI, REST, GraphQL, pytest

AI and machine learning | Claude API, MCP, RAG, LangChain, LlamaIndex, scikit-learn, TensorFlow, Keras

Mobile development | Flutter, Dart, Provider, Firebase, Supabase, SQLite, Material Design

Education

BS Computer Science | University of Central Punjab | Apr 2025 | CGPA 3.78

GitHub Project Portfolio

37 public repositories | 28 distinct projects | 18 completed projects

Data Engineering and Analytics | 7 completed / 7 projects

AI and Machine Learning | 2 completed / 5 projects

Flutter and Mobile Development | 5 completed / 10 projects

Web and Python Applications | 3 completed / 4 projects

Cryptography | 1 completed / 2 projects

Counts use one primary field per project. Completed means a usable implementation or finished case study. Practice, profile and companion repositories do not add to project totals.

Data Engineering Projects

Cloud Data Integration Platform GitHub

Standardizes REST and SOAP ingestion with Azure Functions, Blob Storage, Databricks transformations and data quality checks.

Azure Data Warehouse Ingestion Platform GitHub

Captures commerce API data in ADLS Gen2 and supports retry tracking, transactional PostgreSQL warehouse rebuilds and Power BI reporting.

Construction Budget Data Pipeline GitHub

Automates budget CSV export, schema cleanup and bulk SQL loading for reporting.

Klaviyo SFTP Data Pipeline GitHub

Maps customer CSV fields to a destination schema and delivers validated files through SFTP with an audit log.

Operations Document Automation GitHub

Collects work tickets, cloud documents and staff planning exports into a repeatable reporting workflow.

Freelance Opportunity Intelligence Pipeline GitHub

Collects and enriches listings, matches opportunities to team skills and routes AI-assisted drafts to a human review queue.

AI and Flutter Projects

Marrow Insight GitHub

Built a bone marrow image classification prototype with a Siamese network, a FastAPI prediction service and a Flutter interface.

AskDocAI GitHub

Built a Flutter app for uploading documents, asking questions and generating summaries and quizzes. Used Provider and Firebase Authentication.

ASD Support App GitHub

Built a Flutter app with a FastAPI backend for machine learning predictions and deployed the backend on Vercel.

EsperFlow GitHub

Built donor and recipient workflows for blood donation requests with Firebase authentication, data storage and notifications.

Real-Time Object Detection and Translation

Built a YOLO prototype for video inference and an English-to-Spanish translation prototype exposed through Flask.

Certifications

Supervised Machine Learning: Regression and Classification | DeepLearning.AI and Stanford Online | Apr 2025 Verify

Flutter and Dart: Developing iOS, Android, and Mobile Apps | IBM | Mar 2025 Verify

Getting Started with Flutter and Dart | Packt | May 2025 Verify

AI Basics and Tools for Creativity | University of Michigan | Dec 2024 Verify

Introduction to FastAPI Framework | Duke University | 2025 Verify

Introduction to Large Language Models | Google Cloud | 2023 Verify

Data Collection and Processing with Python | University of Michigan | 2023 Verify

Leadership and Achievements

Management Head | Microsoft Learn Student Ambassadors, UCP | Oct 2024 to Jun 2025
Coordinated chapter members and organized workshops and student events.

Event Organizer | Taakra Speed Programming, UCP | Feb 2025
Organized a programming competition for 50+ participants and coordinated judging and event logistics.

Top 10 Finalist | SOFTEC Capture the Flag | Apr 2025
Placed in the top 10 among 50 teams.

Languages

Urdu (native), English (intermediate)

GitHub Project Directory

Data Engineering and Analytics

Cloud Data Integration Platform | Completed GitHub
Azure Data Warehouse Ingestion Platform | Completed GitHub
Construction Budget Data Pipeline | Completed GitHub
Klaviyo SFTP Data Pipeline | Completed GitHub
Operations Document Automation | Completed GitHub
Freelance Opportunity Intelligence Pipeline | Completed GitHub
Product Analytics Instrumentation Platform | Completed GitHub

AI and Machine Learning

Marrow Insight | Prototype GitHub
ASD Screening Risk Classification | Completed GitHub
HousePrice Estimator | Completed GitHub
SahiCheck | Prototype GitHub
DocuMind | Prototype GitHub

Flutter and Mobile Development

AskDocAI | Completed GitHub
ASD Support App | Completed GitHub
EsperFlow | Completed GitHub
ConvertEasy | Completed GitHub
PDF Viewer | Completed GitHub
AIFile | Prototype GitHub
AR Wardrobe | Prototype GitHub
HomeCircle | In progress GitHub
HomeCircle UI Design System | Companion GitHub
UCP StudyHub | Private project
StudyAce AI | In progress · private

Web and Python Applications

Learning Log | Completed GitHub
Bangels | Completed GitHub
Abdul Hadi Jalil Portfolio | Completed GitHub
Korvexa | In progress GitHub

Cryptography

Hybrid Cryptosystem | Completed GitHub
AES Visualizer | Private project

Practice Coursework and Supporting Repositories

CODESOFT GitHub
LeetCode GitHub
C-Language GitHub
UCP-OS-Labs-with-solutions GitHub
Flutter-image_picker GitHub
coffee-UI-dribble-design-clone GitHub
instagram-ui-clone GitHub
minimalist-tiktok-ui-clone GitHub
dribble-mental-health-ui-clone GitHub
dribble-medical-app-ui GitHub
Abdul-Hadi-Jalil GitHub

Private Supporting Repositories

The private AquaScope source, ASD model and Siamese network repositories support the related projects above.